

M.Tech Project Plan

Project Title: Match Outcome Prediction in Cricket using ML/DL

Problem Statement: Predict win probability of a cricket team dynamically during a match using ball-by-ball and contextual features.

Objectives:

- Develop a pipeline to process ball-by-ball data.
- Engineer context-aware features (RR, RRR, wickets, partnerships).
- Compare ML (LogReg, XGBoost) vs DL (LSTM/GRU).
- Calibrate predicted probabilities for reliability.
- Build an interactive dashboard for visualization.

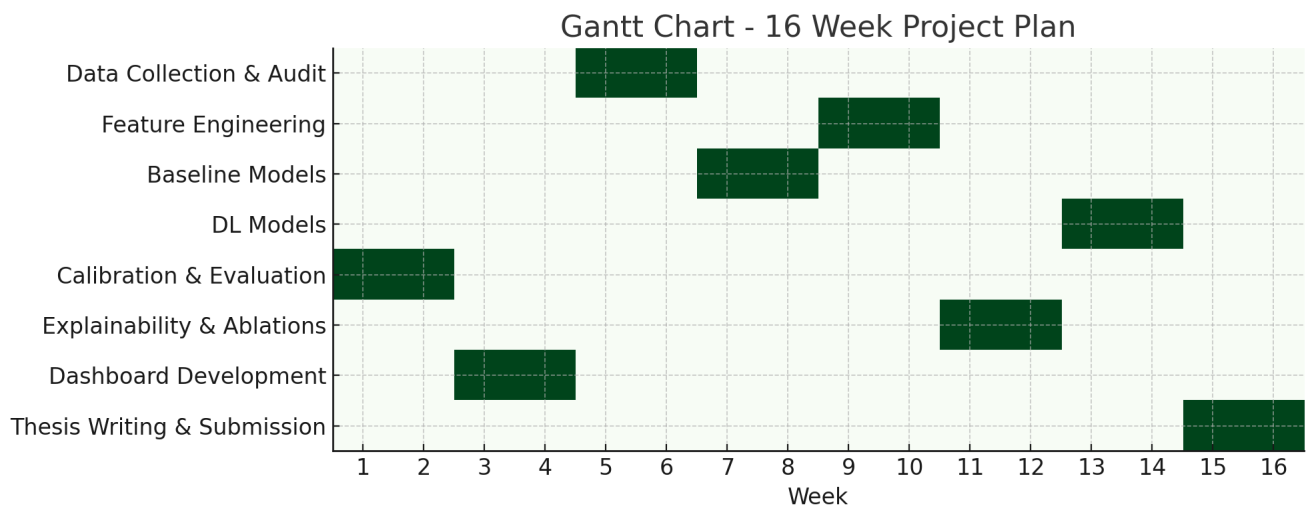
Datasets: Cricsheet, Kaggle IPL/T20 datasets, ESPN Cricinfo stats.

Tools: Python, Pandas, Scikit-learn, XGBoost, PyTorch, Streamlit, MLflow, Plotly.

Chapter Outline:

- 1. Introduction
- 2. Literature Review
- 3. Data & Feature Engineering
- 4. Methodology (ML/DL, calibration, evaluation)
- 5. Experiments & Results
- 6. Dashboard & System Architecture
- 7. Conclusion & Future Work

Project Timeline (Gantt Chart):



Checklist:

- ✓ Dataset collection and preprocessing
- ✓ Feature engineering
- ✓ Baseline ML models
- ✓ Deep learning models
- ✓ Calibration & evaluation
- ✓ Explainability (SHAP, reliability curves)
- ✓ Dashboard implementation
- ✓ Thesis documentation